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Management of Functional Somatic Syndromes and Bodily Distress

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Abstract
Functional somatic syndromes (FSS), like irritable bowel syndrome or fibromyalgia and other symptoms reflecting bodily distress, are common in practically all areas of medicine worldwide. Diagnostic and therapeutic approaches to these symptoms and syndromes vary substantially across and within medical specialties from biomedicine to psychiatry. Patients may become frustrated with the lack of effective treatment, doctors may experience these disorders as difficult to treat, and this type of health problem forms an important component of the global burden of disease. This review intends to develop a unifying perspective on the understanding and management of FSS and bodily distress. Firstly, we present the clinical problem and review current concepts for classification. Secondly, we propose an integrated etiological model which encompasses a wide range of biopsychosocial vulnerability and triggering factors and considers consecutive aggravating and maintaining factors. Thirdly, we systematically scrutinize the current evidence base in terms of an umbrella review of systematic reviews from 2007 to 2017 and give recommendations for treatment for all levels of care, concentrating on developments over the last 10 years. We conclude that activating, patient-involving, and centrally acting therapies appear to be more effective than passive ones that primarily act on peripheral physiology, and we recommend stepped care approaches that translate a truly biopsychosocial approach into actual management of the patient.

Introduction

Functional somatic syndromes (FSS) are well-recognized clusters of bodily symptoms that are common in medical practice and can cause considerable disability. Some, such as irritable bowel syndrome (IBS) and fibromyalgia, are clearly attributed to a single organ system. Some assume a specific etiology such as in the electrosensitivity syndrome. Others are purely descriptive, such as chronic fatigue syndrome. The umbrella concept of FSS was introduced by Barsky and Borus [1] in 1999. It was based on earlier work by Robert Kellner [2] on "psychosomatic syndromes" and included a list of speciality-specific functional syndromes which was only marginally

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supplemented afterwards. However, many patients present with bodily symptoms that do not fit the description of one of these syndromes; although they do not match the pattern of symptoms typical of organic disease yet, those symptoms cause considerable distress and disability. We refer to these presentations as "bodily distress," as a preferable term to "medically unexplained symptoms," which has been widely criticized. Bodily distress refers to persistent bodily symptoms, which are burdensome to the sufferer and usually lead to medical consultations. Bodily distress is common in practically all areas of medicine worldwide [3, 4]. Diagnostic and therapeutic approaches to these symptoms and syndromes vary substantially across and within medical specialties from biomedicine to psychology. However, unifying concepts which are designed to reflect these "psychosomatic" phenomena like the "Diagnostic Criteria for Psychosomatic Research" are rarely applied [5].

Patients may become frustrated with the lack of effective treatment, doctors may experience these disorders as difficult to treat, and this type of health problem forms an important component of the global burden of disease [6]. In this review, we develop a unifying perspective on the understanding and management of FSS and bodily distress. We give an update on current concepts for etiology, classification, and evidence-based recommendations for treatment, concentrating on developments over the last 10 years (for the period before then, see Henningsen et al. [7]).

The Clinical Problem: A Matter of Perspective

Patients present frequently to doctors with bodily symptoms such as pain, palpitations, dizziness, diarrhea, limb weakness, and general fatigue [2]. Some patients suffer from a single, persisting symptom, but others complain of multiple symptoms concurrently, and, over time, the symptoms may or may not fulfill the criteria of an FSS. Such bodily symptoms are often accompanied by psychobehavioral features such as high health anxiety and bodily checking behaviors. The spectrum of severity is wide, ranging from mild symptoms with little functional impairment to severely disabling conditions [2, 8].

In most patients with FSS and bodily distress, there is no well-defined structural organic pathology to be found that correlates to the symptoms. If such pathology is present it does not explain the extent of bodily symptoms and suffering, and even successful treatment and/or remission of the underlying pathology does not relieve the symptoms. In diagnostic follow-up studies, only 0.5% of initial diagnoses of FSS had to be changed into a diagnosis of definite medical disease, whereas an initial thorough evaluation of patients said to have FSS reveals underlying organic pathology in up to 8% of cases [9, 10]. Importantly, the overall number of bodily symptoms is a strong predictor of disability and health care use and appears to be a better measure of overall severity than the severity of a single symptom or single FSS or the description of bodily symptoms as "medically unexplained" [11–13]. High health anxiety is an additional, independent predictor of health care use [14, 15]. Impaired quality of life and work participation in the FSS are at least as severe as those found in well-defined medical diseases with comparable symptoms [16, 17]. Long-term outcome of FSS/bodily distress is variable, but for patients with multiple bodily symptoms it is poor with high rates of disability and sick leave [18]. Notably, low back and neck pain, which have at least large overlap with FSS/bodily distress, are among the top 5 in the global ranking of disability-adjusted life years [19].

In terms of comorbidity, FSS (e.g., IBS and fibromyalgia) and bodily distress in general are associated with higher rates of depression and anxiety than diseases with comparable symptoms attributable to well-defined organic pathology (e.g., inflammatory bowel disease vs. rheumatoid arthritis) [17]. However, many cases of FSS also occur without anxiety or depression; thus, the association between FSS and psychiatric disorder can neither be regarded as a psychological reaction to the bodily symptoms nor as a manifestation of masked or somatized depression or anxiety [20]. Beyond anxiety and depression, personality disorders are a frequent comorbid condition especially in more severe FSS and bodily distress [21].

The prevalence of and the clinical approach to FSS and bodily distress vary according to the clinical context. In general practice, up to 25% of patients are seen with nonspecific bodily symptoms which never receive a specific diagnosis and which often resolve spontaneously. In somatic specialty clinics, up to 50% of patients have such bodily symptoms and are diagnosed as functional symptoms or syndromes often after negative investigations for possible organic disease. These patients often receive treatment focused on the most prominent bodily symptom [7]. Few of these patients reach mental health services (many refuse to attend such clinics), but those who do will usually be diagnosed and treated according to the relevant psychiatric diagnoses: commonly anxiety, depression, or somatization disorder (or "somatic symptom disorder" [SSD]).

Terminology, Classification, and Classificatory Overlap

This multitude of clinical perspectives goes hand in hand with a multitude of descriptive terms, classificatory strategies, and publication traditions in this field, and there is no generally accepted overarching term. The term "medically unexplained symptoms" has been criticized because of its conceptual limitations and practical difficulties in defining insufficient explanation of symptoms [22]. Recently, the term "persistent physical symptoms" has become more popular because of its relatively good acceptance by patients [23], but this is achieved through its nonspecificity. The term "bodily distress" better describes the fact that patients indeed suffer from their bodily symptoms, but for some, "distress" implies a psychological component to this primarily bodily condition, which some patients do not accept.

In terms of classificatory strategies, FSS are usually diagnosed in the different medical sections of the 10th Revision of the International Classification of Diseases (ICD-10) [24], e.g., IBS in the chapter on gastroenterological diseases and fibromyalgia in the chapter on rheumatological diseases. A few of the FSS have research diagnostic criteria, which help to define them more strictly, e.g., the Rome criteria for IBS [25] or the criteria of the American College of Rheumatologists for fibromyalgia [26]. Most diagnoses of FSS include no indication of severity other than a duration criterion for inclusion.

Classification of this group of patients from a mental health perspective has been subject to major changes in recent years. In ICD-10, the classification as a "somatoform disorder" is still based on the absence of an organic medical illness that explains the bodily symptoms. In the 5th edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-5)* [27] of the American Psychiatric Association, the "somatoform disorder" has been replaced (in 2013) by the category of SSD. The latter dropped the "medically unexplained" criterion of DSM-IV somatization disorder but included positive psychobehavioral criteria, namely high health anxiety, excessive symptom preoccupation, excessive health worry, and maladaptive illness behavior. In DSM-5, patients with high health anxiety but no significant bodily symptoms are now categorized as "illness anxiety disorder"[27, 28].

SSD is a new diagnosis and has yet to be tested fully although a self-report instrument to assess its psychobehavioral features has already been published [29]. Abandoning a diagnosis based on the negative criterion of "medically unexplained symptoms" in DSM-5 SSD has been welcomed largely, but the single diagnosis covering all patients with bodily symptoms has been criticized as overinclusive as have the rather arbitrary selection of the positive psychobehavioral criteria [30]. The upcoming revision of ICD (ICD-11) will likely contain a category of "bodily distress disorder" (BDD) that resembles the central characteristics of DSM-5 SSD with emphasis on distressing bodily symptoms and psychobehavioral features [31]. The primary health care section of ICD-11 currently proposes a category of "bodily stress syndrome" (BSS) as replacement for the prior term "medically unexplained somatic complaints." The definition of this new category, despite similarity in name, is rather different from the one of "BDD" in the mental health section [32]: it is based on the presence of ≥3 distressing and disabling somatic symptoms not explained by known physical pathology, without emphasis on psychobehavioral features, and hence on the criteria for "bodily distress syndrome" as elaborated by Fink and Schröder [33]. Hypochondriasis has been retained but will be classified under obsessive-compulsive and related disorders in ICD-11.

These are new diagnostic criteria that have not yet been tested empirically. Both the DSM-5 and ICD-11 mental health diagnoses attempt to define these disorders in a positive manner that emphasizes the severity of bodily symptoms and their accompanying features in a way that (a) distinguishes the disorder from the mild bodily symptoms that occur in healthy people and (b) predict a poor outcome in terms of impaired function or high health care costs [34]. The relationship between the diagnosis of an FSS (e.g., IBS and fibromyalgia) and the mental health diagnostic categories of SSD and BDD is not clear, but considerable overlap is to be expected. Approximately half of the people with a functional somatic syndrome have multiple somatic symptoms beyond the ones required for their specific FSS diagnosis and occurring in many bodily systems. The presence of these multiple bodily symptoms is associated with greater impairment [12]. Similarly, the co-occurrence of an anxiety or depressive disorder with FSS is also associated with greater impairment [17].

Different FSS co-occur in 1 patient more often than would be expected by chance; a finding which has been interpreted as suggesting that the different FSS are actually manifestations of a single disorder [35, 36]. An alternative view is to regard them as separate syndromes which frequently co-occur because of 2 underlying dimensions, which have separate genetic and environmental components: an affective component (depression and anxiety) and a sensory component (especially chronic widespread pain) [37, 38].

Statistical analyses of the common functional symptoms show clusters of cardiopulmonary, gastrointestinal, musculoskeletal, and fatigue symptoms, but the most consistent feature is the division of individuals into those with few or multiple somatic symptoms [39–41]. Two to three percent of the population have multiple FSS; such individuals have more severe illness, more concurrent anxiety or depressive disorders, and greater impairment [13, 42, 43].

To conclude, there is not one single diagnostic category that covers all relevant dimensions for patients with FSS and/or bodily distress. A single FSS diagnosis, like IBS, may be adequate and acceptable for patients with few and specific gastrointestinal symptoms and without significant psychobehavioral features (e.g., health anxiety). The efficacy of different forms of treatment for patients with a single FSS has been tested repeatedly. For those patients who have numerous bodily symptoms relating to several organ systems, ≥1 FSS concurrently, and/or accompanying psychobehavioral features (so-called "complicated bodily distress"), the DSM-5 SSD diagnosis or, in the future, ICD-11 BDD, is more appropriate. These patients with complicated bodily distress require different treatments from those used in patients with a single somatic syndrome.

Etiology

Etiological descriptions of FSS and bodily distress should be based on adequate models of the interplay of peripheral, central, and contextual factors in the experience of bodily distress. Current models of this interplay primarily imply bottom-up processes whereby input from peripheral nociceptive and other sensors is amplified by central or psychosocial factors like sensitization, anxiety, or attribution, with feedback of these central processes impacting on further peripheral input. However, these models cannot explain all phenomena observed in patients with bodily distress well, like their consistently worse accuracy in interoception; they may also underestimate the therapeutic relevance of directly influencing the interoceptive dysfunctions in these patients [44]. More recently, a model of bodily distress as a disorder of perception is gaining ground, where interoception like all perception is seen as being codetermined from scratch by top-down processes, i.e., the expectations or probabilistic predictions, which the central nervous system (CNS) is constantly constructing of its environment including bodily states. According to this model, peripheral input from nociceptors and other sensory structures in the body is only processed bottom up if it does not match the predictions. In the model, which is based on a view of the CNS as a predictive coding machine, disorders of interoception can arise, e.g., when – as a primary failure of inference – overly precise top-down predictions of distress meet with low precision bottom-up input. This then forms the basis for the well-known secondary failure of inferring a symptom of organic illness [44–47]. Such a model grounds CNS-based processes of belief and expectation firmly already in the basic mechanisms of interoception; this also stresses the direct importance of communicative modifications of these processes for the immediate perception of bodily distress.

Applying such a model, some etiological factors will preferentially act either via these top-down or via bottom-up routes, but most imply both routes (see upper and lower parts of Figure 1, called "sensory input" and "expectation"). It is important that, from a clinical point of view, aggravating and maintaining factors (right-hand end of Fig. 1) are more relevant than predisposing and triggering ones, as it is chronic and disabling bodily symptoms which usually require treatment, whereas milder symptoms often resolve spontaneously. In terms of predisposing and triggering factors, it is relevant to consider that bodily distress may be triggered by a viral infection, or news of serious illness in others, or other life stressors, but such events are widespread, and only a small minority of individuals develop bodily distress; vulnerability or host factors must also be involved.

To name exemplary biological and psychosocial examples, genetic factors contribute to the vulnerability for FSS/bodily distress, as well as for chronic pain in general, but only to a limited extent, explaining up to 30% of the variance – epigenetic mechanisms most likely are at least as important here [38, 48]. There is evidence for a genetic tendency to develop bodily distress distinct from that for anxiety and depression [49, 50].

As a psychosocial factor, adverse childhood experiences have an established role as predisposing for FSS, including pseudoneurological syndromes – they raise the odds for the development of FSS up to fourfold, possibly acting via epigenetic mechanisms [51–53]. Personality factors like avoidant and anxious attachment patterns and deficits in emotion regulation have also been linked as predisposing factors to the different facets of bodily distress [54–57]. On another level, cultural factors contribute to the predisposition for bodily distress, with some cultures showing more somatizing tendencies and disability due to bodily distress independent of individual and group level or health care system factors [58, 59].

Acute organic illnesses, stressful work conditions and adverse life events are important precipitating factors for

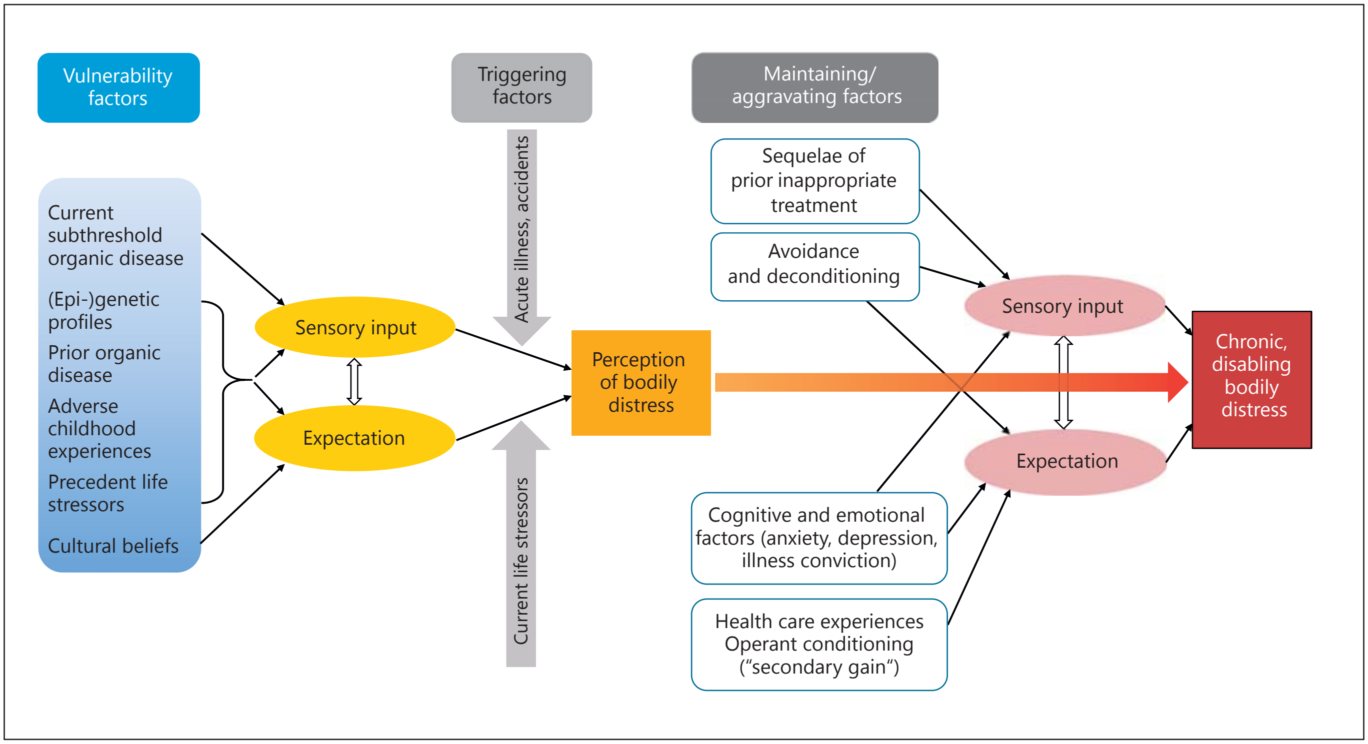

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Fig. 1. Schematic model of the etiology of bodily distress. Note: The distinction of vulnerability/triggering and aggravating/maintaining factors is to some extent artificial as most factors have an influence on both sides.

FSS and bodily distress [60–64]. If persisting, these factors together with the effect of predisposing personality aspects and other important psychosocial state and trait characteristics like health anxiety, illness convictions, and avoidance behavior induce a shift to chronification and contribute to the maintenance of the symptoms of FSS and bodily distress. Further aggravating and maintaining factors arise from the often difficult interactions of these patients with the health care system, leading to missed or late correct diagnosis, inappropriate treatments, and frustrations on all sides. Somatizing communication behavior and persistent beliefs about biomedical causations held by patients and doctors alike, but also systemic factors of the health care system, e.g. reimbursement structures, contribute to these significant barriers for more effective diagnosis and treatment [65, 66].

Management of FSS and Bodily Distress

The 10 years since the last review have seen considerable efforts to aggregate evidence and develop evidence-based recommendations for the management of patients with FSS and bodily distress. These include several national guidelines and Cochrane reviews for bodily distress and many systematic and Cochrane reviews for single FSS. These reviews identified the moderate benefits of various treatments, but they also highlighted the *unmet* treatment needs of this large group of patients by describing the barriers to better diagnosis and treatment [2, 66–71]. We focus here on the management approach to the single patient, but the systemic public health aspects of this clinical problem and the reduction of barriers to better management at this systemic level are also of great importance [2].

Good management of this group of patients should avoid the traps of entrenched dualistic "either mental-or physical" thinking. The patient's bodily symptoms must be taken seriously by the doctor from the outset even though investigations for possible organic disease tend to focus the attention of patient and doctor on the possible general medical diseases that might cause the symptom(s). As part of this balanced "mental as well as physical" approach, the doctor should ask about the whole pattern of bodily symptoms together with symptoms of depression and anxiety as these frequently accompany the bodily symptoms. Similarly, the doctor should explore the patient's beliefs about the cause of the symptoms as a way of identifying and understanding the patient's symptom-related cognitions and emotions, including health anxiety and disease conviction. Good communication with the patient is essential, and, when it is appropriate, the doctor should anticipate the likely outcomes of diagnostic tests, provide positive explanations of the "functional" character of the disorder, and assess the motivation of the patient to actively engage in coping with bodily distress. Stepwise approaches from primary and somatic specialist to integrated care are necessary to adapt to the variations in chronicity and severity, with interventions ranging from watchful waiting to symptomatic relief to multimodal treatment of dysfunctional perceptual and behavioral patterns (see below). Finally, such interventions can be delivered effectively in various medical settings, but seem to be slightly more effective when delivered by mental health care professionals and psychotherapists [72]. Before describing the evidence base for the treatment of FSS and bodily distress, we here give general treatment recommendations and recommendations specific for psychotherapy of these patients.

General Treatment Recommendations for Patients with FSS and Bodily Distress

The following recommendations, though aimed at general practitioners and somatic specialists, form a basis also for interventions of mental health specialists [67, 73].

Assessment

- Think of the possibility of FSS/bodily distress in a patient with persistent physical symptoms; do not equate them with malingering.
- General attitude: slow down, broaden the lens (by asking about the patient's life and circumstances), listen hard.
- Be attentive to clues of the patient indicating bodily or emotional distress beyond the current main symptom and outside your specialist field: screen for other physical symptoms, anxiety, and depression; do not miss medication or alcohol misuse, or suicidal ideations.
- Assess the patient's experiences, expectations, functioning, beliefs and illness behavior, especially with regard to catastrophizing, body checking, avoidance, and dysfunctional health care utilization.
- Avoid repetitive, especially risky investigations only to reassure the patient or yourself.
- Decide whether the patient has mild, moderate, or severe bodily distress/FSS. Are there bodily or mental symptoms clearly beyond a single FSS? Is there excessive loss of functioning? Are there dysfunctional expectations or illness behavior?

Differential Stepped Care

Step 1: Mild FSS/Bodily Distress

- Know yourself and your patient: Ensure an empathetic relationship, use communication skills, pick up the patient's concerns.
- Provide information and reassurance: Explain symptoms, frame tests, discuss test results and your diagnosis in a calming, educating, not trivializing manner.
- Encourage a healthy lifestyle, physical and social activities and distractions, such as sleep hygiene, regular exercising, and fulfilling hobbies.

Step 2: Moderate FSS/Bodily Distress

- Apply measures of step 1.
- Introduce context factors as amplifiers rather than causes for the patient's symptoms. Build an effective, blame-free narrative that is linked to physical as well as psychosocial mechanisms and makes sense to the patient.
- Find and strengthen the patient's individual resources.
- Encourage – and monitor – more functional attitudes and behaviors, such as positive thinking, relaxation techniques, graded exercise, and self-help guides and groups. Set realistic goals together with the patient.
- Provide symptomatic measures such as pain relief or digestives; allow measures from complementary medicine according to the patient's wishes; explain that these measures are temporarily helpful but less effective than self-management.
- Consider antidepressant medication if there is predominant pain or depression.
- If appropriate: set appointments at regular intervals rather than patient initiated.

Step 3: Severe FSS/Bodily Distress

- Apply measures of steps 1 and 2.
- Ensure that traumatic stressors and maintaining context factors, such as domestic violence, medication misuse, factitious symptoms, or litigation, are considered.
- Carefully frame referral to psychotherapist or mental health specialist in addition to reappointment with you.
- Liaise with psychotherapist or mental health specialist on diagnosis, possible difficulties, and further treatment planning.

- When outpatient care is not available or seems insufficient, consider integrated care with multidisciplinary treatment including symptomatic measures, activating physiotherapy, occupational therapy, and psychotherapy.

Initial Steps for Psychotherapy of FSS/Bodily Distress

The following recommendations aim at the initial phases of psychotherapy for patients with at least moderate FSS/bodily distress. They are based on the general recommendations and aim at building a sustainable therapeutic relationship independent of later differentiations according to the pattern of patient problems and school of psychotherapy (adapted and translated from Henningsen and Martin [74]).

- Clarify motivation of the patient for psychotherapeutic consultation. If applicable: confirm to him/her that you acknowledge his/her initial view that the symptoms have an as yet undetected organic basis and that he/she may "only" accept the consultation to please others.
- Use measures from steps 2 and 3 of section Differential Stepped Care.
- Listen attentively to bodily complaints and relationship experiences connected to them (with doctors and other health professionals, with relatives, life partners, colleagues, etc.). Give feedback on the emotional aspects of these experiences (anger, disappointment, fear, etc.).
- In the more chronic patients, give support in organizing the history of presenting complaints (and experiences) into a coherent narrative.
- Encourage the patient to extend his view of the possible influence of psychosocial as well as biological context factors, e.g., through time-limited use of a symptom-context diary (not recommended for patients with very high health anxiety). Do not attempt to "reattribute" symptoms to a predominantly psychosocial cause.
- Negotiate realistic (i.e., modest) treatment goals. Advocate "better adaptation" and "coping," avoid "cure" as a treatment goal.
- Resist the temptation to concentrate on psychosocial issues too early and too independently of lead complaints. If necessary, "somatize", i.e., enquire current bodily symptoms.
- Liaise with others involved in the care of the patient – to obtain relevant information especially concerning the necessity of further somatic diagnostic and therapeutic interventions, but also to send the message to the patient that constructive cooperation in caring for him/her is possible.

Evidence for the Management of FSS and Bodily Distress

To determine the evidence base, we performed an umbrella review of systematic reviews. We searched PubMed, the Cochrane Library, and PsycARTICLES for systematic reviews from January 1, 2007, until September 4, 2017. We used a string of search terms applied in a former review published in 2007 [7]. Additionally, we adapted it to include terms which came up since then [75]. The string of search terms was composed of a set of terms designed for the mental health perspective and for each FSS of interest and combined with a filter specifically designed for each data base (for details of search terms and selection process, see online supplementary material; for all online suppl. material, see www.karger.com/doi/10.1159/000484413).

Terms from the mental health or FSS perspective had to be within the scope of each identified review, and the reviews had to fulfill 3 additional requirements: each review had (1) to define a comprehensive search strategy and (2) a systematic quality evaluation of the included primary studies. (3) The results had to be presented in a structured and differentiated way, allowing the extraction of results for the terms in focus. The review had to report diagnostic criteria used by the primary randomized controlled trials if available. If not, this issue had at least to be addressed. All these criteria were documented systematically in a form designed for this purpose. We did not consider outcomes reported by reviews based on ≤2 primary randomized controlled trials. Reviews targeting children or adolescents were excluded, too. The evidence ratings are based on principles recently developed by the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) Working Group [76]. This approach consists of 2 components: firstly, it considers the magnitude of the observed effect sizes and the precision of conduct and results related to the included primary studies ("quality of evidence"). Secondly, as main innovation, these ratings are amended by an independent estimation of possible adverse effects and information of the cost-benefit/risk ratio for the examined patients ("strength of recommendations"). For the sake of simplicity, we integrated both dimensions into one global rating, termed "composite quality of evidence". It distinguishes 4 grades: *strong*, *moderate*, and *low level of evidence*, and *no evidence* for efficacy of treatment. Most commonly, both rating components corresponded. If not, we voted for a moderate down-grade of potentially harmful but somehow effective interventions. For Cochrane reviews, these GRADE recommendations usually were already determined by the reviews' authors.

In the earlier review, we found it useful for the organization of the evidence on the treatment of FSS and bodily distress to differentiate the following 5 treatment foci and types [7]:

- Peripheral pharmacotherapy primarily aimed at peripheral physiological processes (e.g., bowel function, muscle tension, inflammation, nociceptive pain, etc.).
- Central pharmacotherapy primarily aimed at central processes of sensation, cognition, and affect.
- Active behavioral intervention aimed at changing bodily and interpersonal behaviors, sensations, and cognitions by means of active participation of patients in treatments, such as exercise and different psychotherapies. Multidisciplinary treatments are also included in this group because active behavioral components are essential parts of them.
- Passive physical intervention aimed at passive, nonpharmacological change in peripheral features of the syndrome via physical (including surgical and other skin-penetrating) means.
- For the final treatment group, the rationale is seen as either outside that of those mentioned above (usually a part of complementary or alternative medicine) or is not patient but doctor centered (i.e., aimed at the doctor's behavior via education and training).

In view of the etiological model outlined above, it is now even more evident than 10 years ago that the separation of centrally and peripherally acting and also of activating and passive therapies is largely artificial. Not only will centrally acting agents also have peripheral effects and vice versa, the psychological factors of expectation and attention will also always play a significant part in determining treatment outcomes in seemingly purely passive physical interventions. However, for the purpose of demonstrating the evidence base for treatments of FSS and bodily distress, we decided to keep the separation of these treatment foci as they mirror medical tradition and allow comparison with the former review (Table 1).

In what follows, we discuss some important aspects of the overview of evidence collected in Table 1, where all the relevant references are listed.

Treatment of Bodily Distress

As mentioned above, bodily distress in general as opposed to single FSS is covered by diagnostic categories like SSD that are mostly used in mental health settings. In trials performed under this perspective, primary endpoints refer to functioning, e.g., in measures of health-related quality of life, complementing outcomes in somatic symptom intensity, anxiety, and depression [77]. By the very nature of this perspective and its underlying concept, trials and reviews focusing on peripherally acting drugs and passive physical interventions do not exist. For different forms of short-term psychotherapy and self-help interventions, there is consistently low to moderate evidence for efficacy, as is for consultation letters and psychiatric consultations in primary care. There is no evidence for the efficacy of training in enhanced care for primary care physicians.

Treatment of Single Functional Somatic Syndromes

Most trials reported in systematic reviews since 2007 refer to the treatment of single FSS without stratification according to the total number of bodily symptoms, comorbidity, or other indicators of severity. Primary endpoints often refer to symptom intensity rather than overall functioning. In this group, IBS and fibromyalgia are the functional syndromes with most trials to report, chronic fatigue syndrome is the third of the most prominent FSS, and chronic/nonspecific low-back pain and tension headache/neck pain are not only the ones with many reviews – they also represent the largest global burden of disease in terms of disability-adjusted life years.

- In IBS, most trials refer to drugs that are used to regulate bowel function and hence are categorized as peripherally acting. For most of these components, however, there is only a low or no evidence for efficacy. There is a low level of evidence for the efficacy of antidepressants.
- Various forms of psychological and activating therapies from cognitive behavioral (CBT) to hypnotherapy to mindfulness-based therapy and yoga have demonstrated low to moderate efficacy in treating IBS.
- In fibromyalgia, there are only 3 reviews on peripherally acting agents, demonstrating no evidence for efficacy of nutritional supplements, botulinum toxin, or nonsteroidal anti-inflammatory drugs.
- In contrast, many reviews since 2007 refer to centrally acting agents like antidepressants, pregabalin, or gabapentin. For most of the components in this group, there is only low or no evidence for efficacy. Tricyclic antidepressants still show moderate and hence better evidence for efficacy than newer antidepressants or other agents.
- There are many reviews of trials on a wide range of activating and psychological therapies, including different forms of exercise training and different psychotherapies. For all of these forms of therapy, there is at least low, occasionally moderate, and (for hypnotherapy and multidisciplinary therapy) strong evidence for efficacy.

Table 1. Management of functional somatic syndromes (FSS) and bodily distress (BD) – evidence from systematic reviews since 2007

Type of FSS/BD	Pharmacotherapy		Nonpharmacological therapy		Other (complementary/alternative therapy and doctor centered)
	peripheral	central	psychotherapy and activation	passive physical interventions	
Diagnostic entities for BD in general (multiple unexplained physical symptoms or somatoform/somatic symptom/conversion disorders)	–	New-generation antidepressants + [80©] Tricyclic antidepressants o [80©]	Self-help interventions ++ [81] Psychotherapy + [82©]/++ [83] Short-term psychotherapy + [84, 85©]/++ [86] CBT + [87] Psychological interventions + [88©]	–	Psychiatric consultation in primary care ++ [89] Consultation letters + [90] Enhanced care by generalists o [91©]
Irritable bowel syndrome	Linaclotide (IBS-C) ++ [92–94] Spasmolytic agents + [94–97] Lubiprostone, polyethylene glycol, laxatives (IBS-C) + [94, 98] Loperamide (IBS-D) + [94] Rifaximin + [94, 99, 100] Bulking agents o [97©] Mebeverine o [101]	5-HT3/4 antagonists alosetron + [94, 102–105] (IBS-D)+ [104, 106] Renzapride and cisapride o [101, 107©] (IBS-C) Cilansetron + [103, 104] Antidepressants + [97, 108–111] Tricyclic antidepressants + [108, 111] SSRI + [94]/o [111]	Hypnosis ++ [112] Psychological therapies + [110, 113©, 114]/++ [115, 116] Hypnotherapy + [110, 117©, 118] CBT + [110, 119] Guided self-help, psychoeducational support + [120, 121] Mindfulness-based therapies + [122] Yoga + [123] Minimal-contact psychological treatments + [124]/o [110]	Manipulative therapy + [125]	Chinese herbal medicine + [126] (IBS-D) Probiotics o [127–130]/ + [127, 128, 131–135] Fiber, peppermint oil + [95] Fiber + [136–138] Restricted diet + [139–141] Acupuncture o [142©]/+ [143] Pre- and synbiotics o [133] Lactobacillus + [144]
Fibromyalgia (including myofascial pain/chronic widespread pain)	Botulinum toxin o [145©]/ + [146] Nutritional supplements o [147] Nonsteroidal anti-inflammatory drugs o [148]	Tricyclic antidepressants ++ [149, 150] Pregabalin + [151–155, 156©] SSRI + [149, 150, 157] SNRI + [149]/o [158©] MAOIs + [149, 159©] Duloxetine + [160©, 161] Amitriptyline o [162]/+ [163] Milnacipran o [163, 164©, 165©] Antipsychotic drugs o [166©] Cannabinoids o [167©]	Hypnotherapy +++ [168] Multidisciplinary therapy +++ [169]/ ++ [170, 171] CBT ++ [172, 173©] Psychological therapies ++ [174] Aerobic exercise ++ [175©]/+ [176, 177] Exercise + [178–181] Aquatic exercise + [182©] Resistance exercise training + [183] Community-deliverable exercise + [178] Guided imagery/hypnosis + [184] Mind and body therapy + [147, 184] Bodily awareness interventions + [185] Complementary and alternative exercise + [186, 187] Tai chi, qigong + [188]/o [181, 189, 19033]	Balneo-/hydrotherapy ++ [147, 180, 191–193] Electric stimulation o [180, 194] Repetitive transcranial magnetic stimulation o [180, 195] Massage o [147, 180, 196]	Homeopathy + [197] Acupuncture o [147, 198–200, 201©]/+ [202, 203] Dry needling o [204]
Chronic fatigue syndrome	–	–	CBT ++ [205©, 206] Exercise therapy ++ [207©] Behavioral interventions with a graded physical activity component ++ [208]	–	Acupuncture + [209, 210] Combined moxibustion and acupuncture + [210]
Chronic low back pain	Nonsteroidal anti-inflammatory drugs + [211, 212©] Capsaicin + [213©] Paracetamol o [214©]	Antidepressants o [211, 215] Opioid analgesics, + [211]/ o [216©]	CBT ++ [217] Behavioral treatment + [218©] Exercise therapy + [219, 220] Strength/resistance and coordination/stabilization exercise + [221] Motor control exercise + [222©] Unloaded movement facilitation exercise + [223] Yoga + [224©] Cardiorespiratory and combined exercise o [221] Pilates-based exercise o [225] Muscle energy technique o [226©]	Radiofrequency denervation + [227©] Massage + [228©] Spinal manipulation o [229, 230©, 231] Low-level laser therapy o [232©]/+ [233] Therapeutic ultrasound o [234©] Combined chiropractic interventions o [235©] Transcutaneous electrical nerve stimulation o [236©]	Multidisciplinary biopsychosocial rehabilitation + [237©] Acupuncture o [231]/ + [229, 238] Education o [239©, 240]
Tension headache, chronic neck pain	Dipyrone + [241] Ibuprofen + [242©] Paracetamol + [243©] Ketoprofen + [244©] Aspirin o [245©] Botulinum toxin A o [246]	Tricyclic antidepressants + [247]/o [248] SSRI/SNRI (venlafaxine) o [249©]	CBT ++ [137] Multimodal treatments + [250] Exercise + [250]	Multimodal manual therapy ++ [251] Manual therapy, spinal/thoracic manipulation + [250] Infrared laser o [250]	Acupuncture + [252, 253©] Education o [240]

Table 1 (continued)

Type of FSS/BD	Pharmacotherapy		Nonpharmacological therapy		Other (complementary/alternative therapy and doctor centered)
	peripheral	central	psychotherapy and activation	passive physical interventions	
Temporomandibular joint disorder	Botulinum toxin o [254]	–	Manual therapy and therapeutic exercise + [255] Exercise o [256]	Occlusal stabilization splint + [257–259] Musculoskeletal manual approach + [260] Temporomandibular lavage o [261]	–
Atypical face pain/myofacial pain/burning mouth syndrome/bruxism	Systemic α-lipoic acid + [262] Capsaicin o [262]	Clonazepam + [262, 263] Amitriptyline o [264©]	–	–	–
Non-ulcer dyspepsia/functional dyspepsia	Prokinetics + [265, 266] Acotiamide (acetylcholinesterase inhibitor) o [267] Proton pump inhibitors ++ [268©]	(5-HT)-1A receptor agonists o [269] Mosapride/cisapride o [270, 271] Antidepressants o [272]/+ [269]	–	–	Acupuncture o [273©]
Chronic pelvic pain	α-Blockers + [274] Progesterone + [275©]	–	–	Laparoscopic uterosacral nerve ablation o [276]	Acupuncture + [277]
Interstitial cystitis/painful bladder syndrome	Pentosan polysulfate + [278] Intravesical resin-iferatoxin + [279] Botulinum toxin A + [280, 281] Bacillus Calmette–Guérin + [280] Intravesical treatments o [282©]	–	–	–	–
Repetitive strain injury	–	–	Exercise o [283©]	Neural gliding techniques o [284]	–
Premenstrual syndrome/dysmenorrhea	β2-Adrenoceptor agonists o [285©] Vertix agnus castus + [286] Oral contraceptives o [287©] Noncontraceptive estrogen-containing preparations + [288©]	SSRI + [289©, 290]	CBT + [290] Psychological interventions + [291©]	–	–
Atypical/nonspecific chest pain	–	Antidepressants + [292]	Psychological interventions ++ [293©]	–	–
Tinnitus	Ginkgo biloba o [294] Zinc supplementation o [295©] Anticonvulsants o [296©]	Tricyclic antidepressants o [297©]	CBT + [298©]/++ [299] Self-help intervention + [300]	Repeated transcranial magnetic stimulation o [301©]	Acupuncture o [302, 303]
Dizziness	–	–	Psychotherapy + [304]	–	–

IBS, irritable bowel syndrome; C, constipation; D, diarrhea; SSRI, selective serotonin reuptake inhibitors; SNRI, selective serotonin and norepinephrine reuptake inhibitor; CBT, cognitive behavioral therapy; 5-HT, 5-hydroxytryptamine. We do not list treatments with scarce empirical basis (≤2 source randomized controlled trials per intervention × FSS). For simplicity, strength of evidence for efficacy of a specific treatment type is indicated in 4 different grades, with the reviews contributing to this summary estimate. The ratings represent integrated "composite grades of evidence": +++, strong level; ++, moderate level; +, low level; o, no evidence/recommendation for efficacy of treatment and strength of recommendation; –, no reviews included. "©" denotes Cochrane review (68 Cochrane reviews are included). For such an integration of systematic reviews, which use different criteria as well as an extensive variety of outcomes and represent different opinions in heterogeneous clinical fields, an estimation of effect sizes was not feasible. General empirical trends in FSS management are shown; it is not an adequate basis for individual treatment recommendations. The terms used were taken from the systematic reviews and vary in grade of differentiation (e.g., for some FSS the reviews state the evidence for psychotherapy, whereas for others they state the evidence for different forms of psychotherapy separately).

- Among passive physical interventions, there is only one form with consistently moderate evidence for efficacy in several reviews, balneo-/hydrotherapy (treatment with baths/water).
- In chronic fatigue syndrome, surprisingly few systematic reviews were detected, confirming a trend already apparent in 2007. For both CBT and exercise therapy, there is low to moderate evidence for efficacy. For most other forms of treatments, there are no reviews available since 2007 which fulfill the criteria set defined here.
- In nonspecific chronic low-back pain, there is evidence of low quality for the efficacy of nonsteroidal anti-inflammatory drugs and capsaicin, but not for paracetamol. There is neither evidence for the efficacy of antidepressants nor for the efficacy of most passive physical interventions except for radiofrequency denervation or – different to the fibromyalgia syndrome – for massage.
- For activating therapies and psychotherapy, there is mixed evidence, with some therapies like Pilates-based exercise showing no evidence, whilst other exercise-based interventions demonstrate low evidence, or, finally, CBT showing moderately good evidence for efficacy.
- For tension headache and chronic neck pain, there is only low to no evidence for the efficacy of nonsteroidal analgesics, which, moreover, intend only short-term pain relief.
- For CBT, exercise, and multimodal therapies, there is low to moderate evidence for efficacy. Among the other forms of treatment, a review of multimodal manual therapy shows moderate evidence for efficacy, and 2 reviews of acupuncture show low evidence. Other forms of therapy show no evidence for efficacy.
- The pattern of reviews and the evidence reported therein is heterogeneous for the other FSS in Table 1, and in many syndromes only 1 or 2 types of treatment are documented in systematic reviews. In syndromes like interstitial cystitis or functional dyspepsia, several reviews report trials with peripherally acting agents, with mostly low to no evidence for efficacy. In others like dizziness or tinnitus, different forms of activating therapy and psychotherapies show low or moderate evidence for efficacy.

In summary, the evidence base has evolved considerably but in essence has not changed profoundly since our last review in 2007 [7]. It still documents low to moderate evidence with small to moderate effect sizes, with overall best evidence for activating, patient-involving treatments, i.e., the types of treatment typically offered, among others, by mental health specialists and psychotherapists. For all other types of treatment, the evidence is more mixed and often negative. In view of the overwhelmingly negative results of reviews of trials of passive physical interventions, this type of treatment should require special justification to be tested at all in future trials.

All in all, there is no doubt that there is a need for further broadening of the evidence base in the treatment of patients with bodily distress and FSS.

To obtain larger treatment effects in future trials, the following should be considered:

- In view of the large heterogeneity of patients also within one diagnostic group, patient cohorts also in trials of single FSS should regularly be stratified according to the total number of bodily symptoms and other indicators of severity, and other specifiers like illness-related cognitions should also be taken into account in order to achieve personalized, potentially more effective treatments.
- To enhance comparability, all clinical trials in this heterogeneous field should use a common set of core outcome domains – a European working group recently suggested the following: (1) classification, (2) intensity and interference, (3) associated psychobehavioral features and biological markers, (4) illness consequences (quality of life, disability, health care utilization, and health care costs), (5) global improvement or treatment satisfaction, and (6) unwanted negative effects [78].
- Innovative treatment approaches (e.g., expectation management in early stages of secondary prevention) have to be tested.
- Predictors and mechanisms of change should be investigated (e.g., in dismantling studies).

Conclusions

Stepped care approaches appear to be best suited at all levels of care considering the large spectrum of severity in patients with FSS and bodily distress. Initially and in uncomplicated cases, an encompassing biopsychosocial attitude, a focus on symptomatic relief, patient activation, and avoidance of iatrogenic harm is particularly helpful. In more chronic and/or severe cases, management works best when not only the patients but also their doctors achieve a reframing of the clinical problem: from cure to care and coping, from classical biomedical explanations to a broader view of biologically and psychosocially aggravating and alleviating factors [79]. Importantly, this reframing from cure to care and coping is also necessary for mental health specialists and psychotherapists, as is the switch from classical psychosocial explanations to such a broader view of biopsychosocial modulators.

In addition to these individual-centered measures, more systemic approaches, including under- and postgraduate medical training [80], and adaptations of health care systems to avoid mismanagement of these often chronic and costly patients are urgent.

Disclosure Statement

All authors declare no competing interests.

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